embedUR

Linux Assingment-3

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Scenario: Automating file backup and Reporting to the system. Create a shell script called "backup_manager.sh" that performs the following tasks incorporating the concepts suggested.

Requirements:

1. Command-line Arguments and Quoting:
The script must accept three arguments: Source directory: A directory containing files to back up. Backup directory: The destination where files will be backed up. File extension: A specific file extension to filter (e.g., .txt).
Example: ./backup_manager.sh "/home/user/source" "/backup" ".txt"

2. Globbing:
The script should use globbing to find all files in the source directory matching the provided file extension.

3. Export Statements:
Use export to set an environment variable BACKUP_COUNT, which tracks the total number of files backed up during the script execution.

4. Array Operations:
Store the list of files to be backed up in an array.
Print the names of these files along with their sizes before performing the backup.

5. Conditional Execution:
If the backup directory does not exist, create it. If creation fails, exit with an error.
If the source directory is empty or contains no files matching the extension, exit with a message.
If a file already exists in the backup directory with the same name, only overwrite it if it is older than the source file (compare timestamps).

6. Output Report:
After the backup, generate a summary report displaying:
Total files processed.
Total size of files backed up.
The path to the backup directory.
The report should be saved in the backup directory as backup_report.log.

Command:

```
mkdir source backup
echo “Hello File 1” > source/file1.txt
echo “Hello File 2” > source/file2.txt
echo “Hi and Hello” > source/file3.txt
nano backup_manager.sh
chmod +x backup_manager.sh
./backup_manager.sh source backup txt
```

Code:

```
#!/bin/bash


SOURCE_DIR="$1"

BACKUP_DIR="$2"

EXT="$3"


export BACKUP_COUNT=0


if [ ! -d "$SOURCE_DIR" ]; then

   echo "Source directory does not exist"

   exit 1

fi


if [ ! -d "$BACKUP_DIR" ]; then

   mkdir "$BACKUP_DIR"

   if [ $? -ne 0 ]; then

      echo "Failed to create backup directory"

      exit 1

   fi

fi


FILES=("$SOURCE_DIR"/*."$EXT")
```

```
if [ ! -e "${FILES[0]}" ]; then
   echo "No files with .$EXT found"
   exit 0
fi

echo "Files to be backed up:"
for file in "${FILES[@]}"; do
   echo "$(basename "$file") - $(stat -c %s "$file") bytes"
done

for file in "${FILES[@]}"; do
   filename=$(basename "$file")

   if [ -f "$BACKUP_DIR/$filename" ]; then
      if [ "$file" -nt "$BACKUP_DIR/$filename" ]; then
         cp "$file" "$BACKUP_DIR/"
         BACKUP_COUNT=$((BACKUP_COUNT + 1))
      fi
   else
      cp "$file" "$BACKUP_DIR/"
      BACKUP_COUNT=$((BACKUP_COUNT + 1))
   fi
done

REPORT="$BACKUP_DIR/backup_report.log"

echo "Backup Summary" > "$REPORT"
echo "-----------------" >> "$REPORT"
echo "Total files backed up: $BACKUP_COUNT" >> "$REPORT"
echo "Backup directory: $BACKUP_DIR" >> "$REPORT"
```

```
Feb 15 17:58
vboxuser@Ubuntu: ~/source
vboxuser@Ubuntu:~/source$ mkdir source backup
vboxuser@Ubuntu:~/source$ echo "Hello File 1" > source/file1.txt
vboxuser@Ubuntu:~/source$ echo "Hello File 2" > source/file2.txt
vboxuser@Ubuntu:~/source$ echo "Hi and Hello" > source/file3.txt
vboxuser@Ubuntu:~/source$ nano backup_manager.sh
vboxuser@Ubuntu:~/source$ nano backup_manager.sh
vboxuser@Ubuntu:~/source$ ./backup_manager.sh source backup txt
bash: ./backup_manager.sh: Permission denied
vboxuser@Ubuntu:~/source$ chmod +x backup
backup/             backup_manager.sh
vboxuser@Ubuntu:~/source$ chmod +x backup_manager.sh
vboxuser@Ubuntu:~/source$ ./backup_manager.sh source backup txt
Files to be backed up:
file1.txt - 13 bytes
file2.txt - 13 bytes
file3.txt - 13 bytes
vboxuser@Ubuntu:~/source$
```

```
Feb 15 17:59
vboxuser@Ubuntu: ~/source
  GNU nano 7.2                    backup_manager.sh
#!/bin/bash
shopt -s nullglob
SOURCE_DIR="$1"
BACKUP_DIR="$2"
EXT="$3"
export BACKUP_COUNT=0

if [ ! -d "$SOURCE_DIR" ]; then
    echo "Source directory does not exist"
    exit 1
fi

if [ ! -d "$BACKUP_DIR" ]; then
    mkdir "$BACKUP_DIR"
    if [ $? -ne 0 ]; then
        echo "Failed to create backup directory"
        exit 1
    fi
fi

FILES=("$SOURCE_DIR"/*."$EXT")

if [ ! -e "${FILES[0]}" ]; then
    echo "No files with .$EXT found"
    exit 0
fi

echo "Files to be backed up:"

^G Help      ^O Write Out  ^W Where Is   ^K Cut       ^T Execute    ^C Location   M-U Undo      M-A Set Mark
^X Exit      ^R Read File  ^\ Replace    ^U Paste     ^J Justify    ^/ Go To Line M-E Redo      M-6 Copy
```

Feb 15 17:59
vboxuser@Ubuntu: ~/source
GNU nano 7.2
backup_manager.sh
echo "Files to be backed up:"
for file in "${FILES[@]}"; do
echo "$(basename "$file") - $(stat -c %s "$file") bytes"
done
for file in "${FILES[@]}"; do
filename=$(basename "$file")
if [-f "$BACKUP_DIR/$filename"]; then
if ["$file" -nt "$BACKUP_DIR/$filename"]; then
cp "$file" "$BACKUP_DIR/"
BACKUP_COUNT=$((BACKUP_COUNT + 1))
fi
else
cp "$file" "$BACKUP_DIR/"
BACKUP_COUNT=$((BACKUP_COUNT + 1))
fi
done
REPORT="$BACKUP_DIR/backup_report.log"
echo "Backup Summary" > "$REPORT"
echo "----------------" >> "$REPORT"
echo "Total files backed up: $BACKUP_COUNT" >> "$REPORT"
echo "Backup directory: $BACKUP_DIR" >> "$REPORT"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location M-U Undo M-A Set Mark
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line M-E Redo M-6 Copy